

Academic Supervisor's Feedback on Thesis

(Research project)

Completed by a Student in Year 4, group DSBA211 of the Bachelor's Programme "Data Science and Business Analytics" at the HSE University Faculty of Computer Science

Evelina Victorovna Urusova

Full Student name

on the topic: "Implementing Machine Learning for Optimizing Debit Product Marketing"

No.	Evaluation Criteria (only assess those applicable to student's work) There is no requirement of research novelty for the BSc student work	Score (on a 10-point scale)	Comments <i>(Here is an approximate list of the competencies developed by the student while completing the thesis. You can refer to them when describing the degree to which the evaluation criteria have been met and place your own comments in the column)</i>
1	Research novelty	9	Evelina does not claim theoretical discovery, yet the thesis extends the state-of-practice in Russian retail banking. She fuses three strands that have never been deployed together at T-Bank: (i) class-transformation uplift inside a high-cardinality CatBoost model, (ii) a Bayesian channel-prior layer refreshed nightly from live A/B streams, and (iii) an eight-day micro-KPI tailored to early-life adoption of a children's debit card. The student formalised the causal objective $\Delta(x)$ with Neyman–Rubin notation, justified class transformation algebraically, and proved the merit through AUUC $\uparrow 94\%$. Competencies: UC-2, UC-3, PC-2, PC-4 fully demonstrated.
2	Relevance of the work	10	The Junior Card is strategically important yet historically under-converting ($\approx 5\%$). Evelina's pipeline doubled the contacted conversion to 10 % while cutting blended CPA by 14 % and e-mail load by 18 %. These outcomes are already visible in March-May live dashboards. The work addresses a concrete pain-point for Marketing, Risk, and Compliance alike, meeting UC-3, PC-2 at the highest level.
3	Comprehensive comparative review of known results	10	Chapter 3 offers a rigorous, bilingual literature survey: 26 English and 11 Russian items, critical comparison of segmentation, classical ML, uplift frameworks, and channel-adaptation papers. The student reproduced key formulae (logistic, SVM, GBM, uplift) and connected them to business constraints. Excellent mastery of UC-5, PC-12, PC-13.
4	Complexity and volume of the completed work	10	• Ingested 480 M multichannel events and 11 Bn telemetry rows. • Engineered 300 + raw features, pruned to 87 via dummy probes $\rightarrow$ Boruta $\rightarrow$ CatBoost SHAP. • Ran two full modelling generations, 5-fold CV, 3 temporal hold-outs, uplift metrics. • Built production-grade stack: Prefect orchestration, nightly Beta–Binomial updater, FastAPI scorer (p99 = 42 ms), Grafana drift board. • Delivered a Flask demo "Junior-Card Uplift Studio" to visualise real-time decisions. This breadth showcases PC-6, PC-8, PC-9 at an industrial standard.

5	Quality of the composed text. Clear and structured presentation of ideas.	9	The 31-page report is clear, logically segmented, formulae and figures properly cross-referenced. English-language style surpasses usual BSc level; Russian abstract is concise and accurate. Minor typographical repetitions persist in Sections 5–6 but do not hinder readability. Competencies PC-12, PC-13, PC-14, PC-15 are strongly met.
6	Adherence to the planned work schedule, meeting the deadlines for completing the main stages of the thesis, interaction with supervisor	10	Evelina delivered weekly artefacts exactly on plan: EDA-week 1, V 1-week 3, V 2-week 6, business impact-week 8, demo app-week 9. Communication was proactive: she proposed extra causal-loss experiments and SHAP-drift alarms. Exhibits outstanding PC-10, PC-17, PC-18.
FINAL SCORE		10	FINAL SCORE 10 / 10 - Excellent - General commentary: Evelina Urusova's thesis fully satisfies - and in most places exceeds - the requirements for a bachelor-level research project. The work is highly relevant to T-Bank's commercial goals, attacking a real pain-point (low Junior-Card uptake, high messaging cost) with a rigorously engineered, data-driven solution. - Depth and complexity: The project spans large-scale data engineering (480 M communication events), advanced feature selection, comparative ML benchmarking, causal uplift modelling, and a demonstrative micro-service. This breadth demonstrates solid mastery of competencies PC-2, PC-6, PC-8, PC-9. - Analytical quality & novelty: The integration of a CatBoost class-transformation uplift model with a nightly-refreshed Bayesian channel layer is original within the bank and delivers clear incremental value. AUUC gains of 0.0016 → 0.0033 and a live out-of-time AUC jump from 0.77 to 0.87 confirm technical soundness. Competencies UC-2, UC-3, PC-4 are convincingly met. - Business impact. After rollout, the contacted eight-day conversion rate increased from 5.2 % to 10.3 %-a $\approx 2\times$ uplift - while blended CPA fell 14 %. Message volume shifted toward push and banners, cutting e-mail load 18 % and reducing unsubscribe complaints. These metrics translate into marketing efficiency and exemplify the student's ability to link analytics with P&L outcomes. - Quality of exposition: The 31-page report is well-structured, bilingual, equation-backed, and figure-rich. Code samples, diagrams, and the lightweight Flask demo ("Junior-Card Uplift Studio") make the solution transparent to both technical and business stakeholders-evidence of PC-12, PC-13, PC-15. - Discipline & collaboration: All milestones were delivered exactly on schedule; communication was proactive and solutions-oriented, reflecting PC-10, PC-17, PC-18.

		Strengths: • Clear causal formulation and thorough literature review. • Robust evaluation on balanced, out-of-time, and out-of-sample splits. • Tangible business metrics demonstrating value. • Production-grade implementation with monitoring and drift safeguards. Areas to watch: • KPI horizon is eight days; future work should validate lifetime value effects. • Nightly prior refresh may miss intra-day shocks-streaming updates planned. Taking into account the technical depth, methodological rigour, and documented business impact, the thesis merits the highest mark of 10.
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Academic Supervisor:

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